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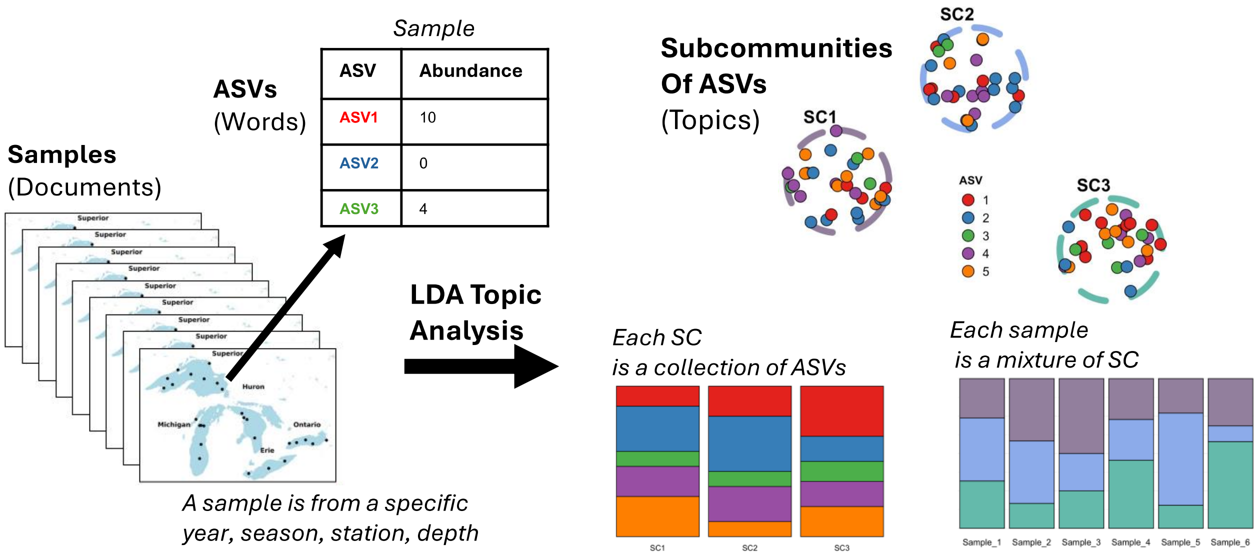

Fig 1. Conceptual diagram illustrating the application of Latent Dirichlet Allocation (LDA) to microbial community data from the Great Lakes. In this framework, each water sample is treated like a document, composed of words (taxa or ASVs) with varying abundances. LDA models the distribution of taxa across samples by identifying topics—interpreted here as microbial subcommunities (SC). Each subcommunity is a probabilistic mixture of taxa, and each sample is modeled as a mixture of these subcommunities or topics. This approach allows for dimensionality reduction and facilitates ecological interpretation of microbial community structure.

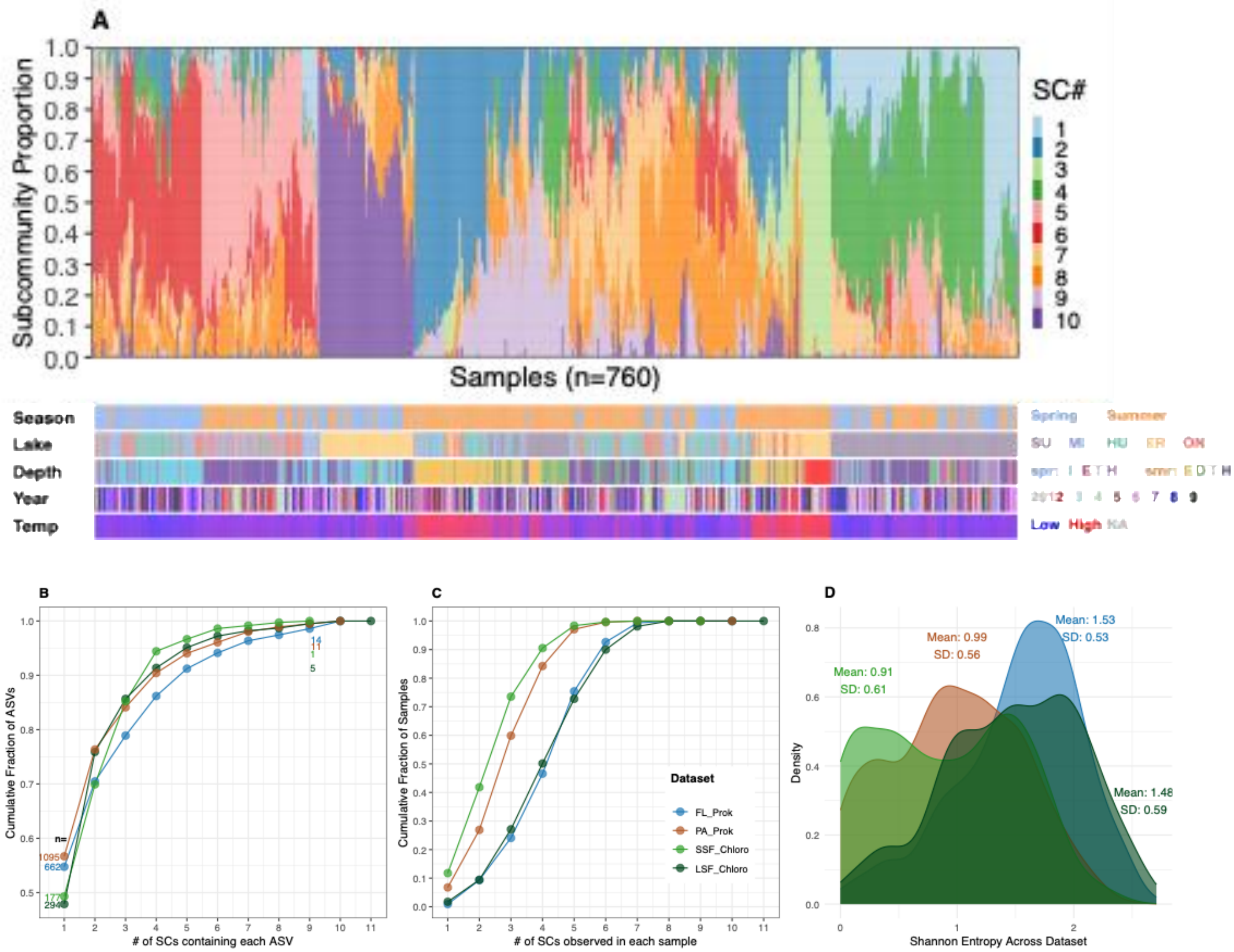

Fig. 2. Subcommunity organization across biological blocks. (A) Sample-by-subcommunity (SC) composition for free-living prokaryotes (FL, n=760), with samples ordered by Ward clustering and colored by SC identity. Metadata tracks below show season, lake, depth, year, and temperature. (B) Cumulative distribution of ASVs by the number of SCs in which they were detected (p>0.0001) across all four size fractions; n values indicate the number of ASVs found in exactly 1 or 9 SCs (the highest number of SCs all datasets have). (C) Cumulative distribution of samples by the number of SCs detected per sample (p>0.01). (D) Shannon entropy of SC composition per sample for each dataset.

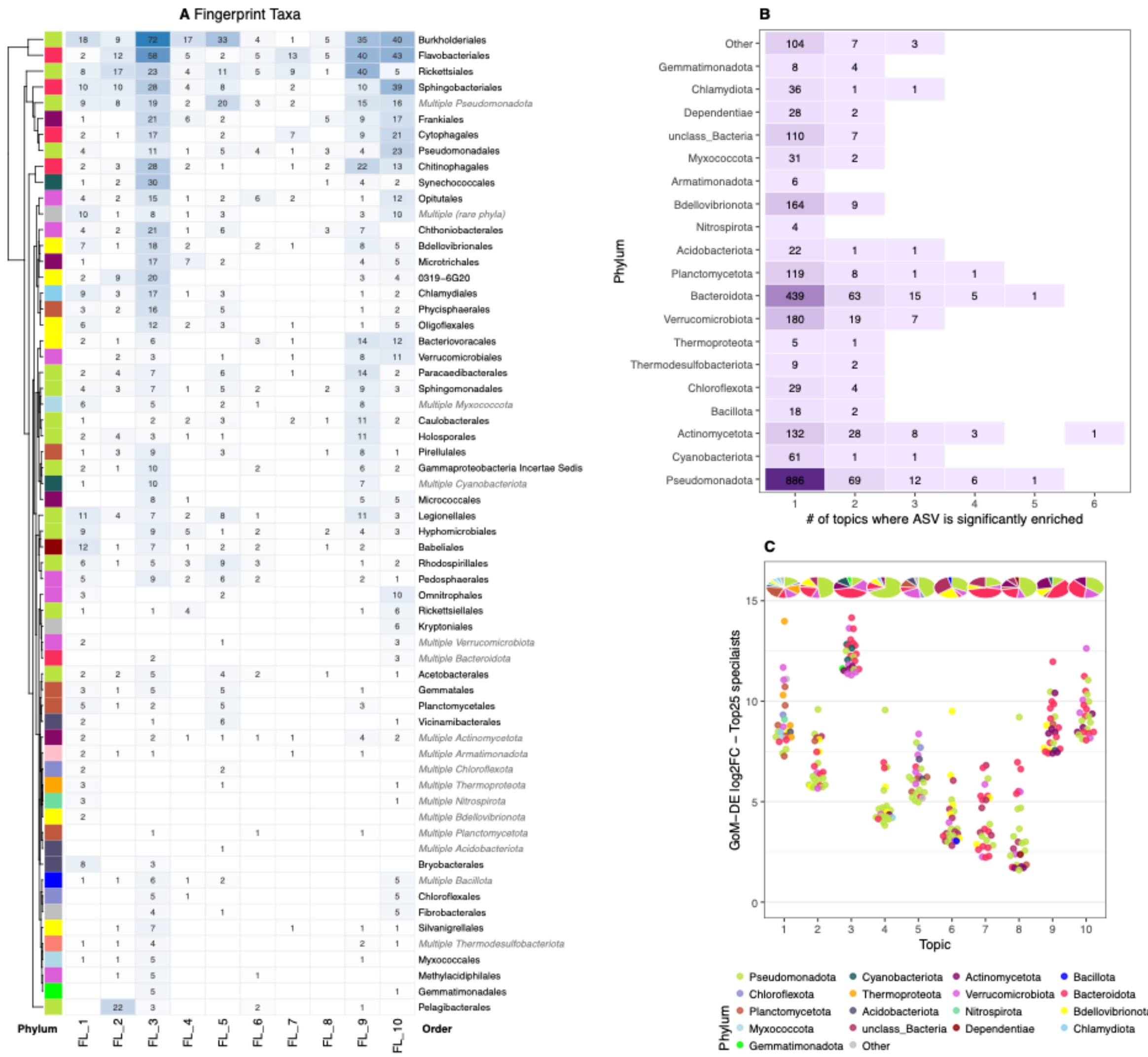

Fig. 3. GoM-DE fingerprint taxa and subcommunity specificity for free-living prokaryotes. (A) Fingerprint taxa heatmap showing the number of differentially enriched ASVs per taxonomic order across all 10 FL subcommunities (SCs), restricted to ASVs enriched in exactly one SC (see panel B). Rows are taxonomic orders clustered by phylum identity (dendrogram); row label colors distinguish named orders (black) from phylum-level collapsed entries (grey italic, e.g., Multiple Pseudomonadota). Values indicate ASV counts per SC. Row/phylum colors correspond to the legend in panel C. Full GoM-DE results for PA, SSF, and LSF are shown in Supplementary Fig. 5-7. (B) Distribution of all marker ASVs (n=2,688) by the number of SCs in which each ASV was significantly enriched. (C) log2 fold-change of the top 25 specialist ASVs per SC. Points are colored by phylum; pie charts above each SC show the phylum composition of all specialist ASVs for that topic.

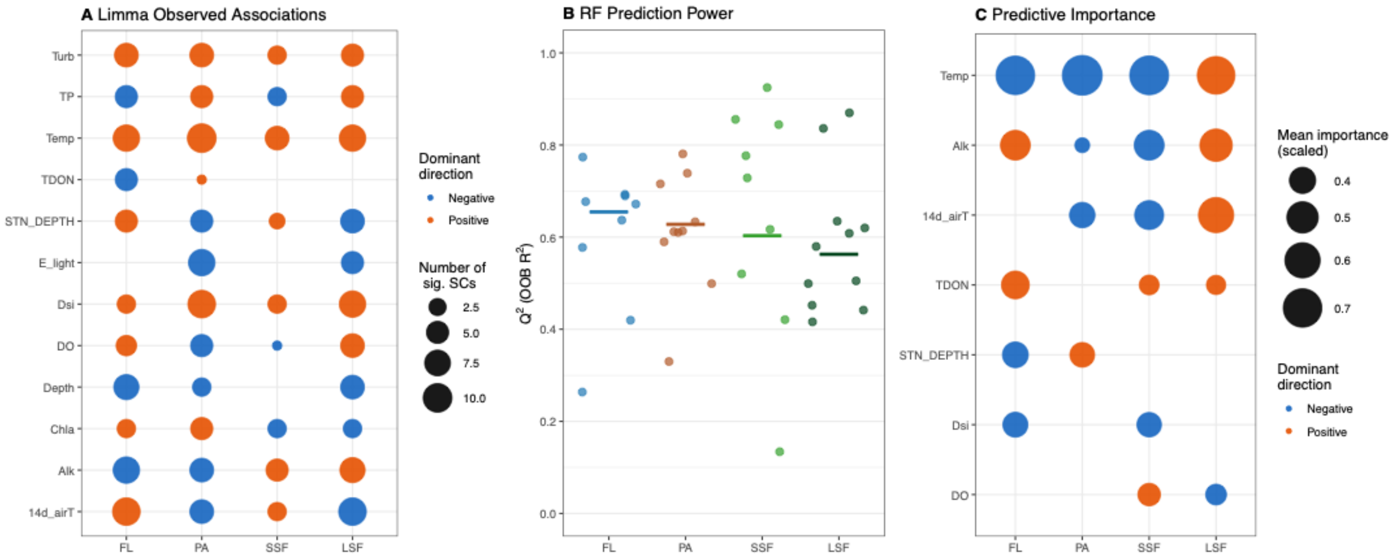

Fig. 4. Environmental structuring of subcommunities across biological blocks. (A) Limma observed associations between environmental variables and SC composition across all four biological blocks. Circle size indicates the number of SCs with a statistically significant association (BH-adjusted $p < 0.05$); color indicates dominant direction of the t-statistic across significant SCs (orange = positive, blue = negative). Full per-SC Limma results for each block are shown in Supplementary Fig. 6–9C. (B) RF prediction power (Q^2; OOB R^2) for each SC across all four biological blocks. Each point represents one SC; horizontal bars indicate block means. (C) Cross-block summary of RF+SHAP predictive importance. Circle size reflects mean scaled SHAP importance across SCs within each block; color indicates dominant direction derived from SHAP value correlations (orange = positive, blue = negative) where at least 60% of SCs within a block agree. Full RF and SHAP results per block are shown in Supplementary Fig. 6–9D.

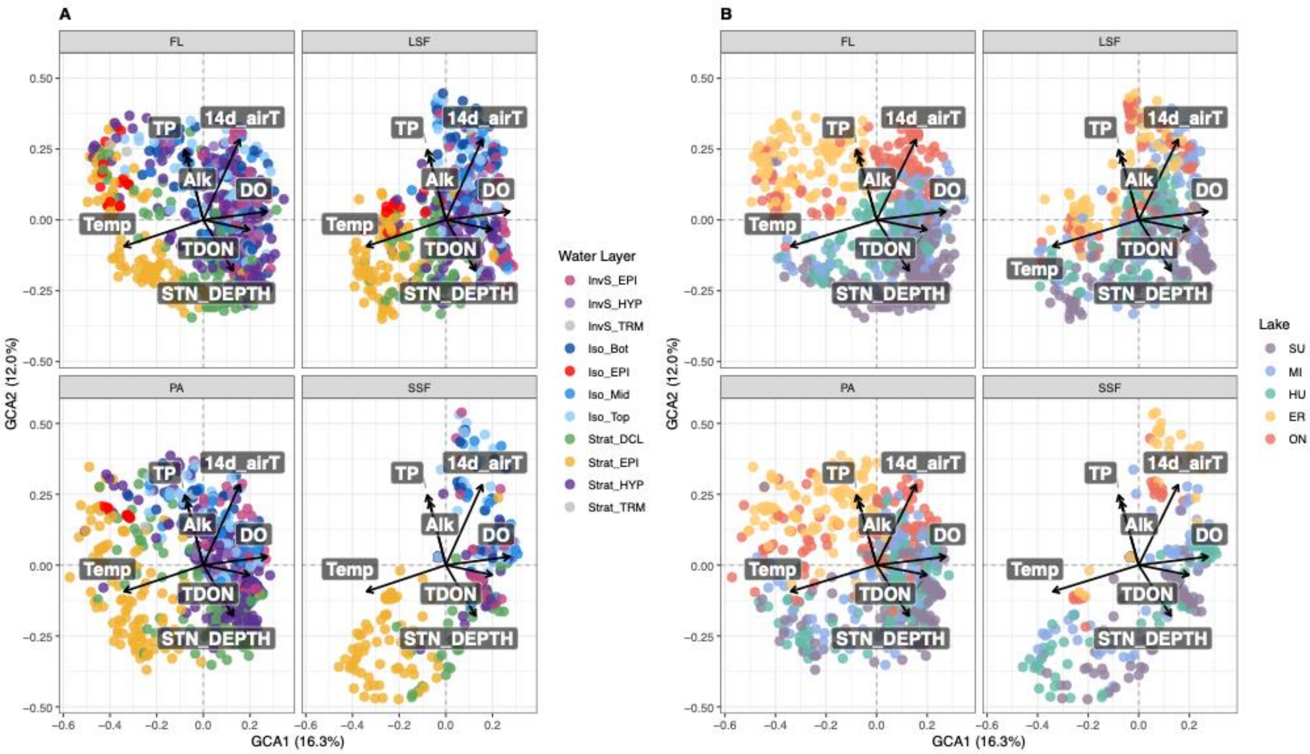

Fig. 5. Group Compositional Analysis (GCA) biplots across all four biological blocks. Sample scores on GCA1 and GCA2 are shown for free-living prokaryotes (FL), particle-associated prokaryotes (PA), small size fraction chloroplasts (SSF), and large size fraction chloroplasts (LSF). Environmental vectors show the direction and relative magnitude of environmental associations with GCA axes; only the longest 80% of vectors are shown for clarity. (A) Samples colored by water layer. (B) Samples colored by lake. Block contributions and environmental loadings for all five axes are shown in Supplementary Fig. 15.

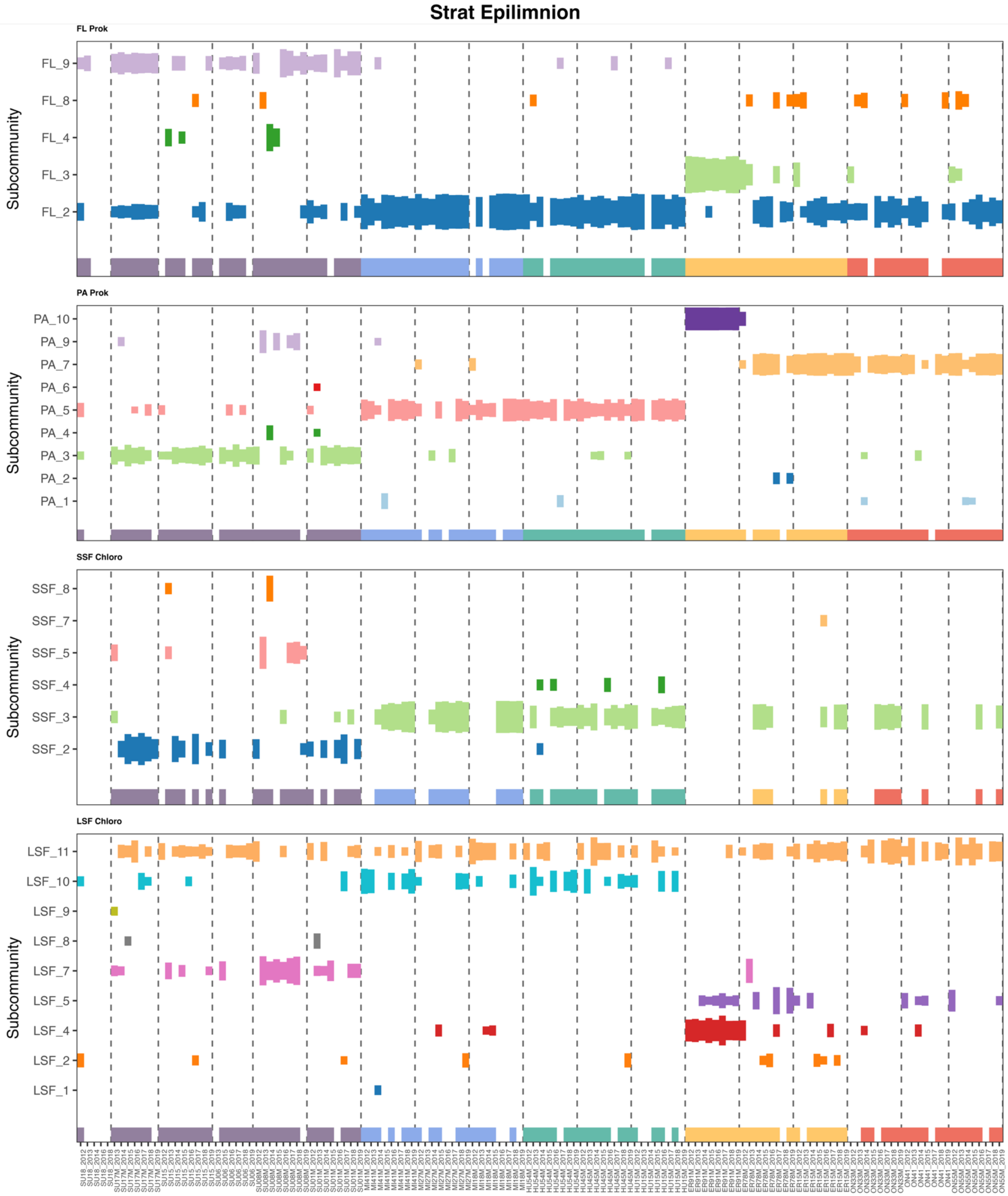

Fig. 6. Subcommunity composition of warm stratified epilimnion waters. SC proportional abundance across samples for all four biological blocks, ordered from western Lake Superior to eastern Lake Ontario stations with years 2012–2019 nested within each station (dashed lines). Only SCs contributing more than three times the uniform expectation (3/K, where K is the number of SCs) are shown.

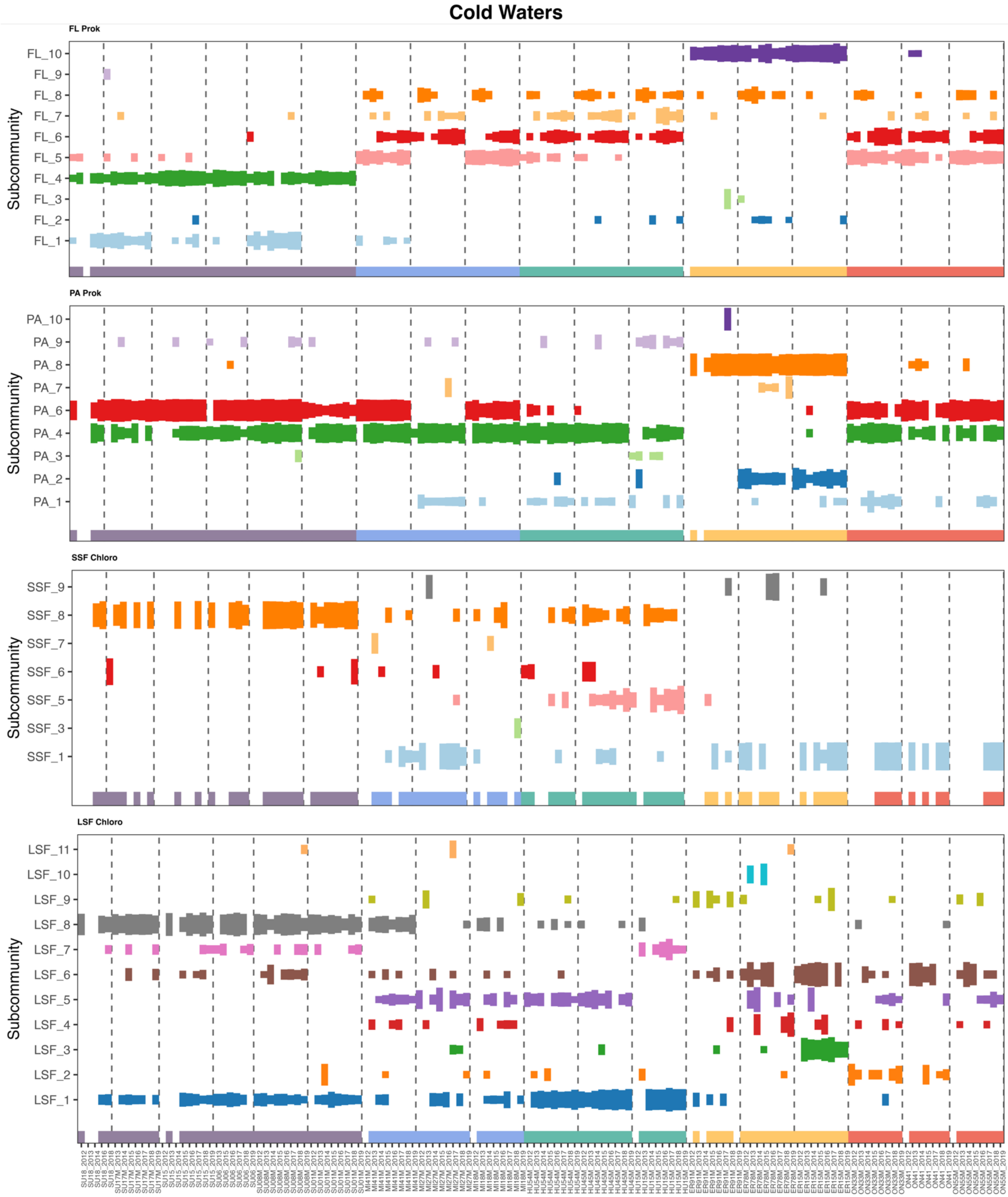

Fig. 7. Subcommunity composition of cold water habitats. SC proportional abundance across samples for all four biological blocks, ordered from western Lake Superior to eastern Lake Ontario stations with years 2012–2019 nested within each station (dashed lines). Cold water samples include stratified hypolimnion, spring isothermal, and spring inversely-stratified conditions. Only SCs contributing more than three times the uniform expectation (3/K, where K is the number of SCs) are shown.

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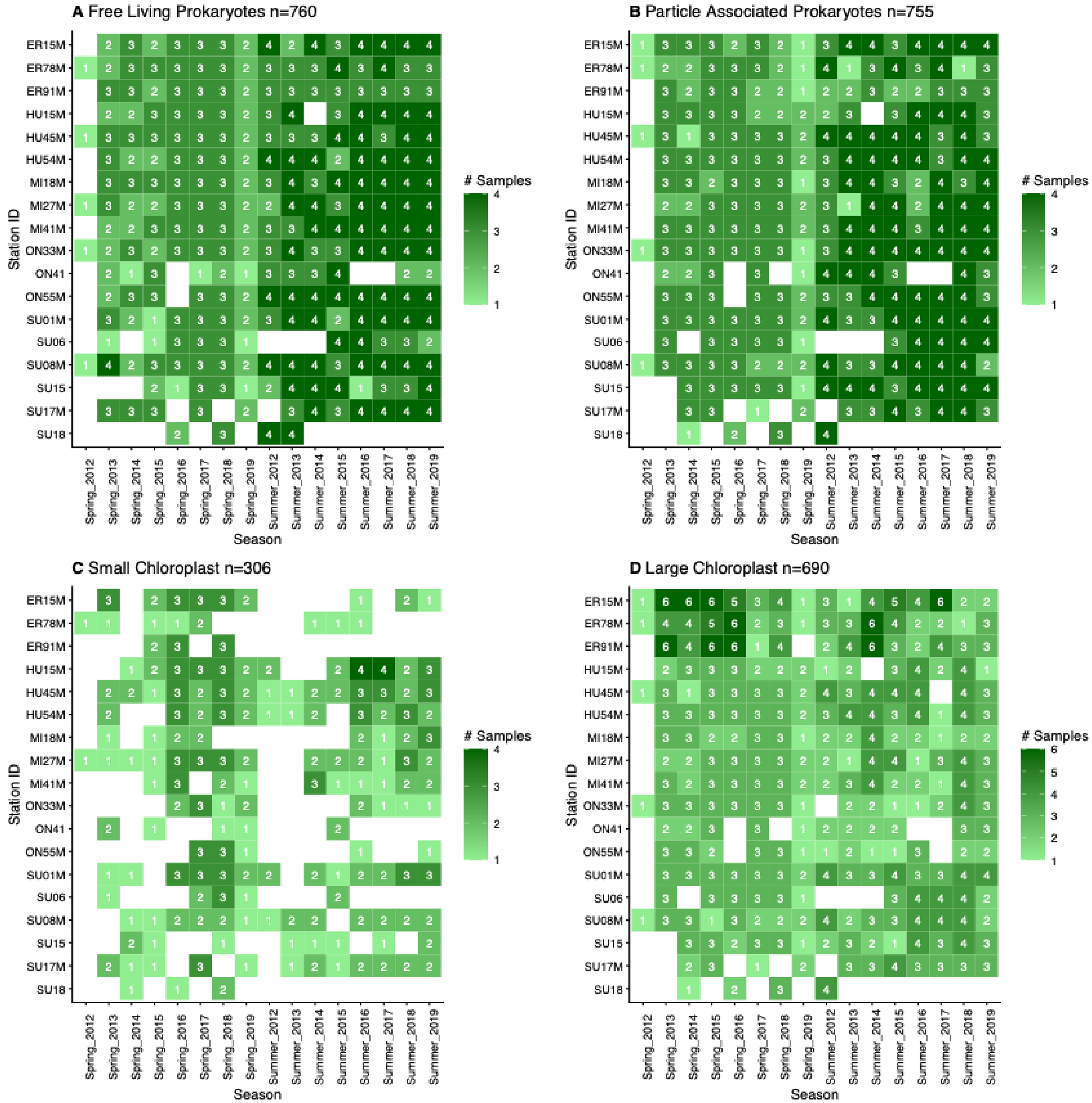
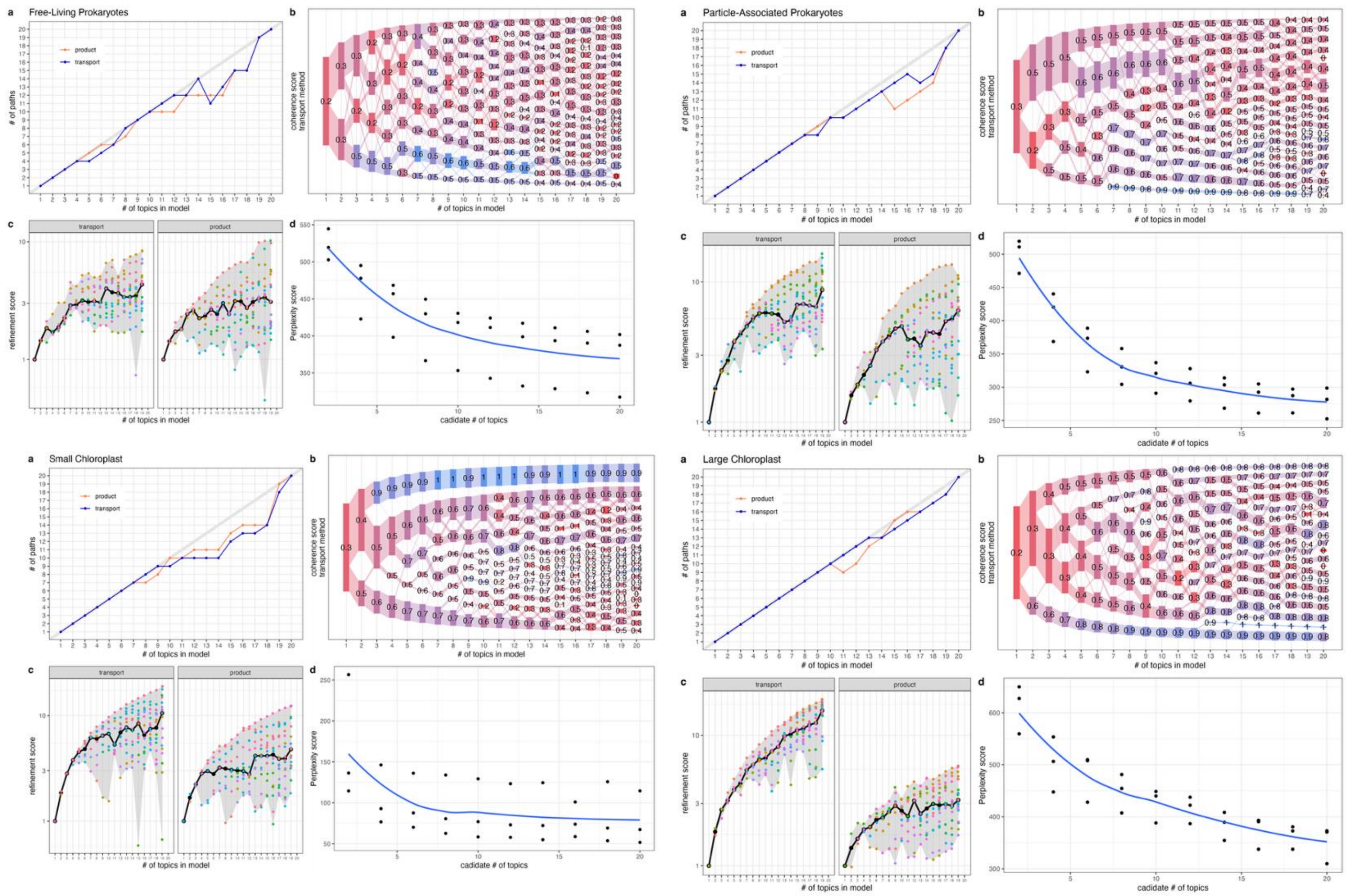
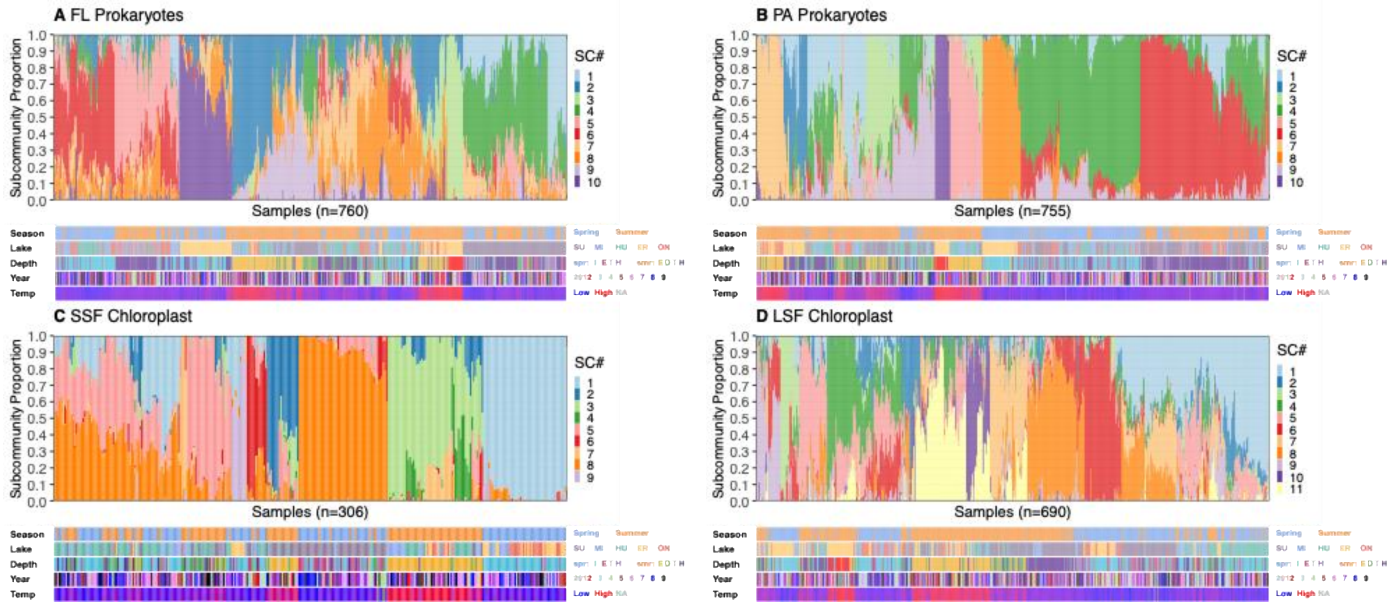


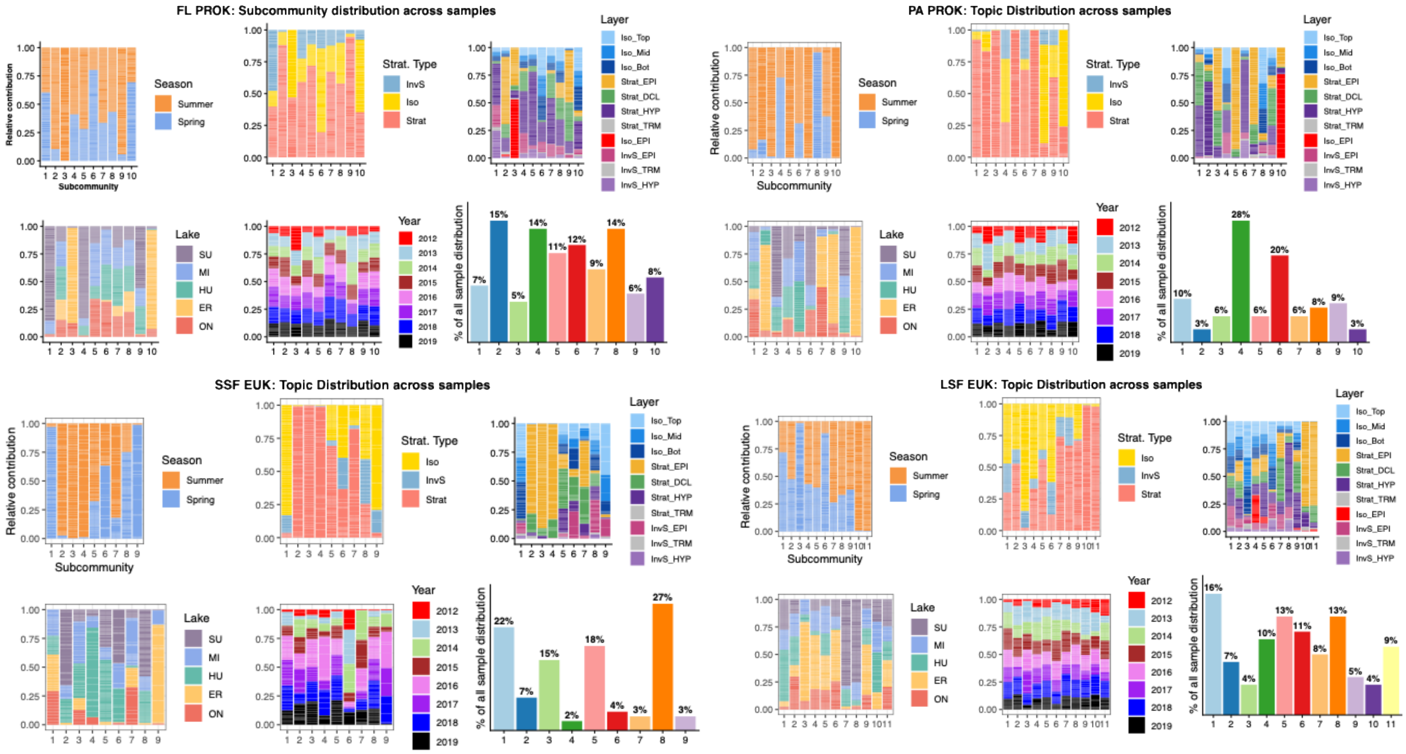
Supp Fig 1. Sampling effort for LDA analysis across years (x-axis) and stations (y-axis) for each dataset: (A) FL_prokaryotes, (B) PA_prokaryotes, (C) SSF_chloroplasts, and (D) LSF_chloroplasts. Color indicates the number of samples per station-year.



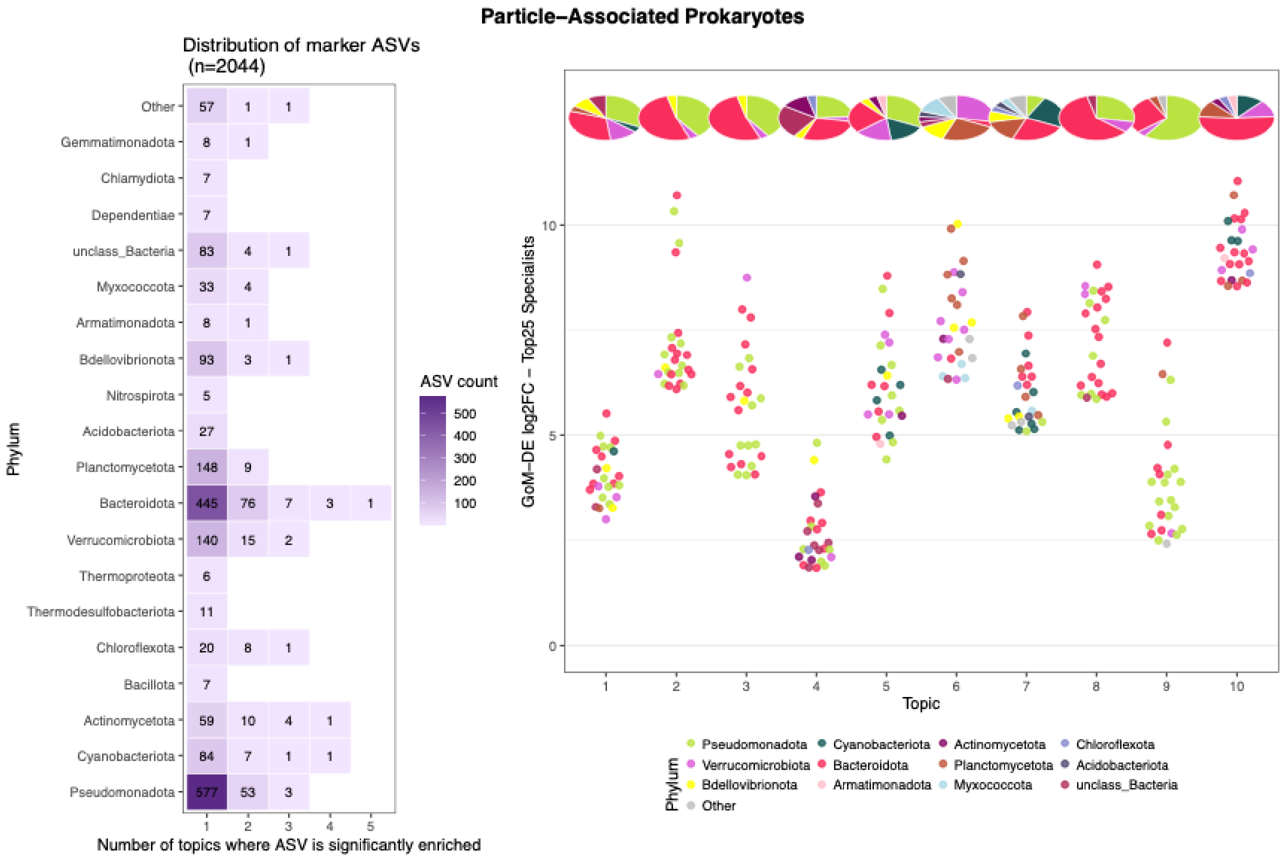
Supp Fig. 2. LDA model diagnostics used to select the number of topics (subcommunities, SCs) for each size fraction. Each panel shows four diagnostic plots: (top left) number of paths, with lines indicating performance from product- and transport-based methods; (bottom left) refinement score for each method; (top right) coherence score; and (bottom right) perplexity.



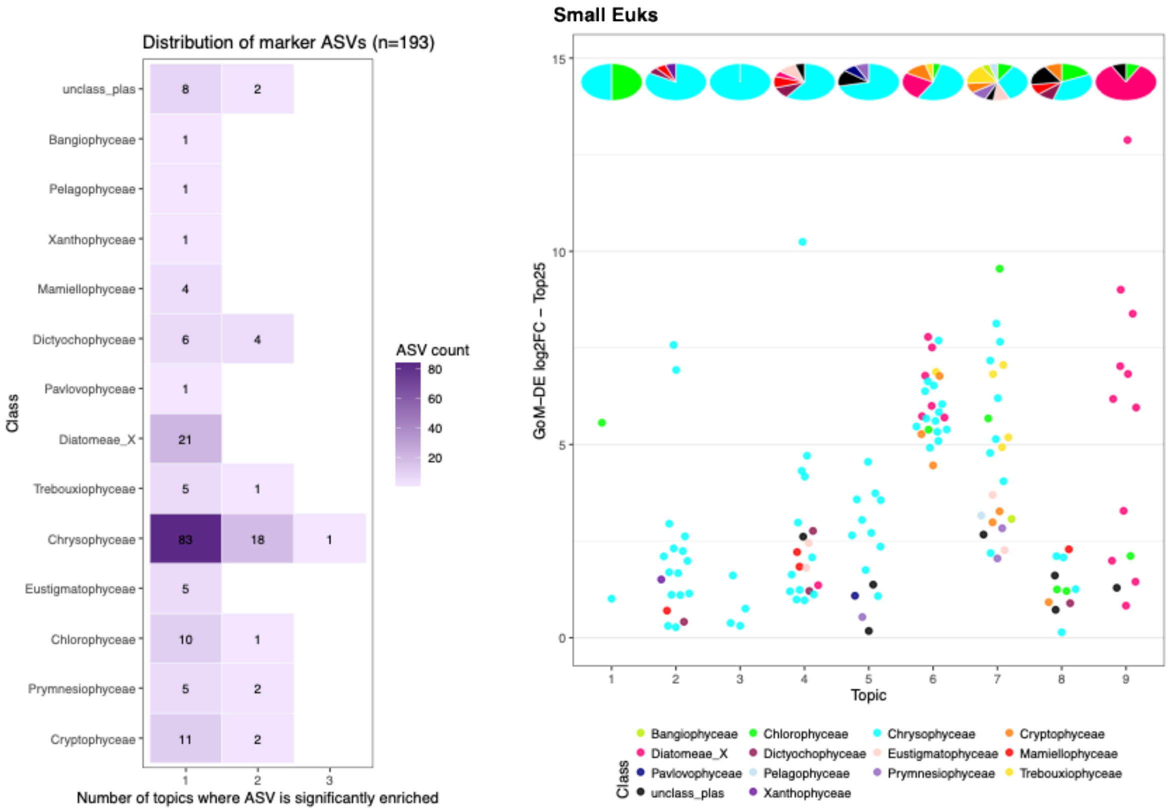
Supp Fig. 3. Sample-by-subcommunity (SC) composition for different blocks with samples ordered by Ward clustering and colored by SC identity. Metadata tracks below show season, lake, depth, year, and temperature.



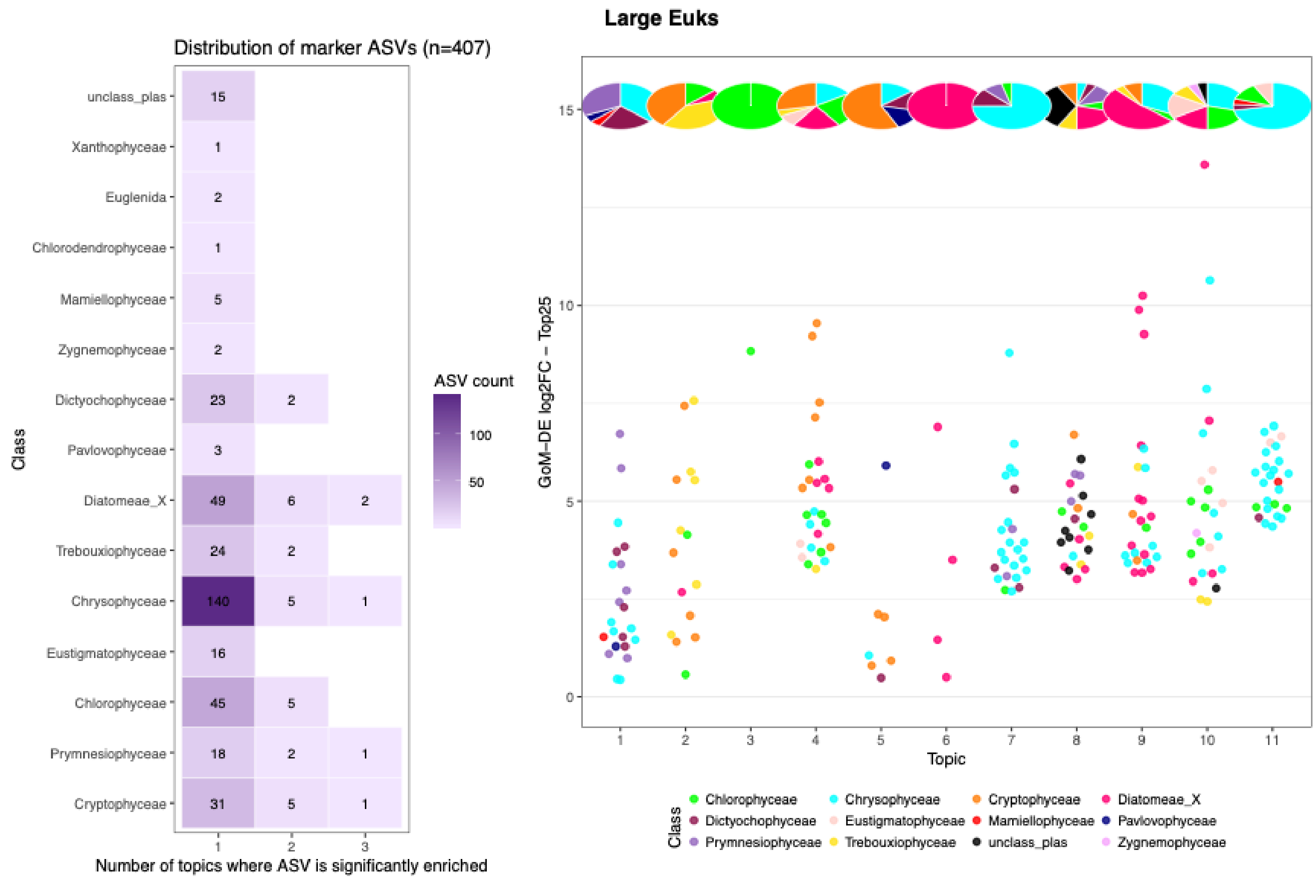
Supp Fig. 4. Environmental and spatial context of microbial subcommunities. Panels show the relative contribution of each SC across samples, grouped by metadata categories including season, stratification type, water column strata, lake, and year. The final bar (far right) shows the percentage of all samples in which each SC was detected, indicating sub-community prevalence across the dataset.



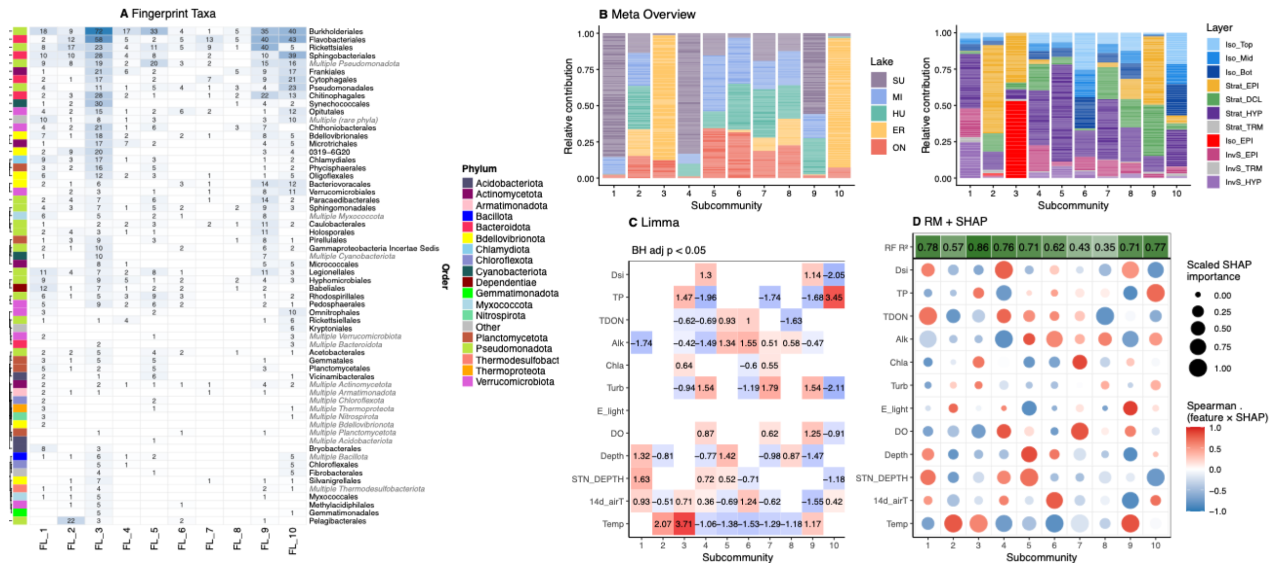
Supp. Fig 5. GoMDE distributions. (left) Distribution of all marker ASVs (n=2,044) by the number of SCs in which each ASV was significantly enriched. (right) log2 fold-change of the top 25 specialist ASVs per SC. Points are colored by phylum; pie charts above each SC show the phylum composition of all specialist ASVs for that topic.



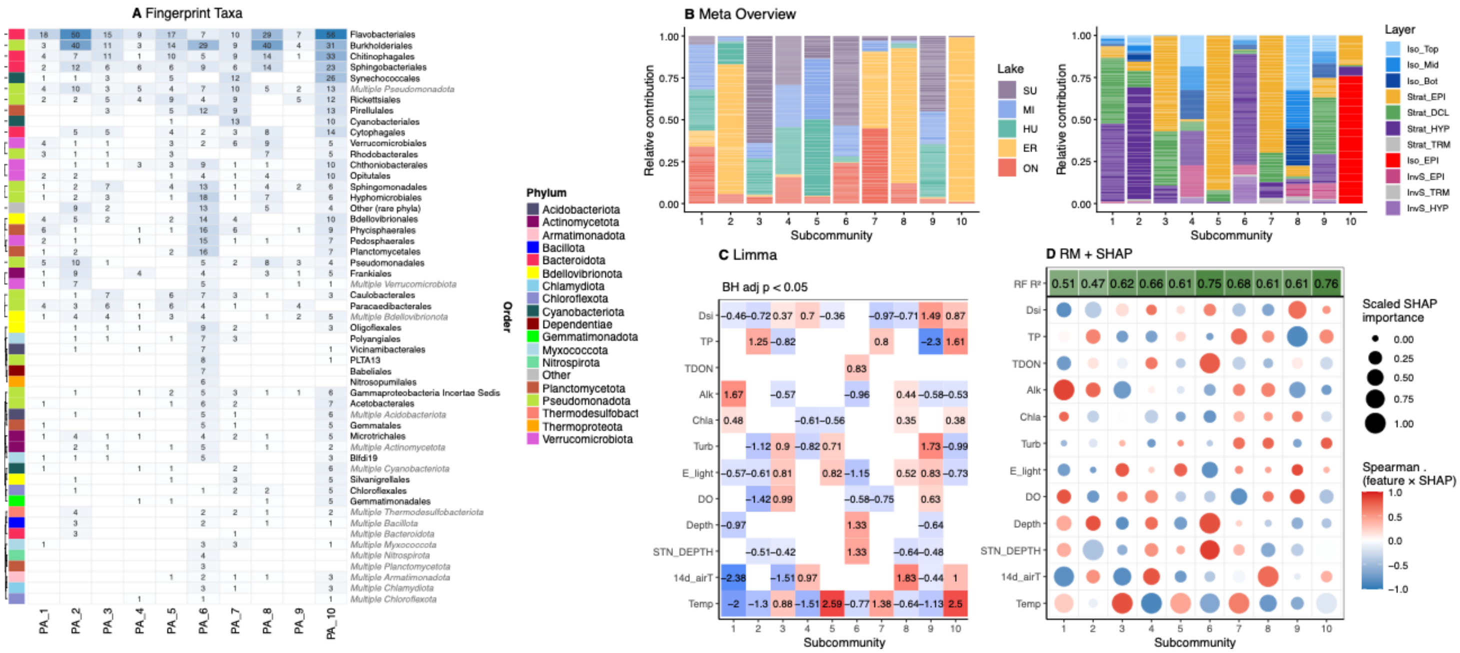
Supp. Fig 6. GoMDE distributions. (left) Distribution of all marker ASVs (n=2,044) by the number of SCs in which each ASV was significantly enriched. (right) log2 fold-change of the top 25 specialist ASVs per SC. Points are colored by phylum; pie charts above each SC show the phylum composition of all specialist ASVs for that topic.



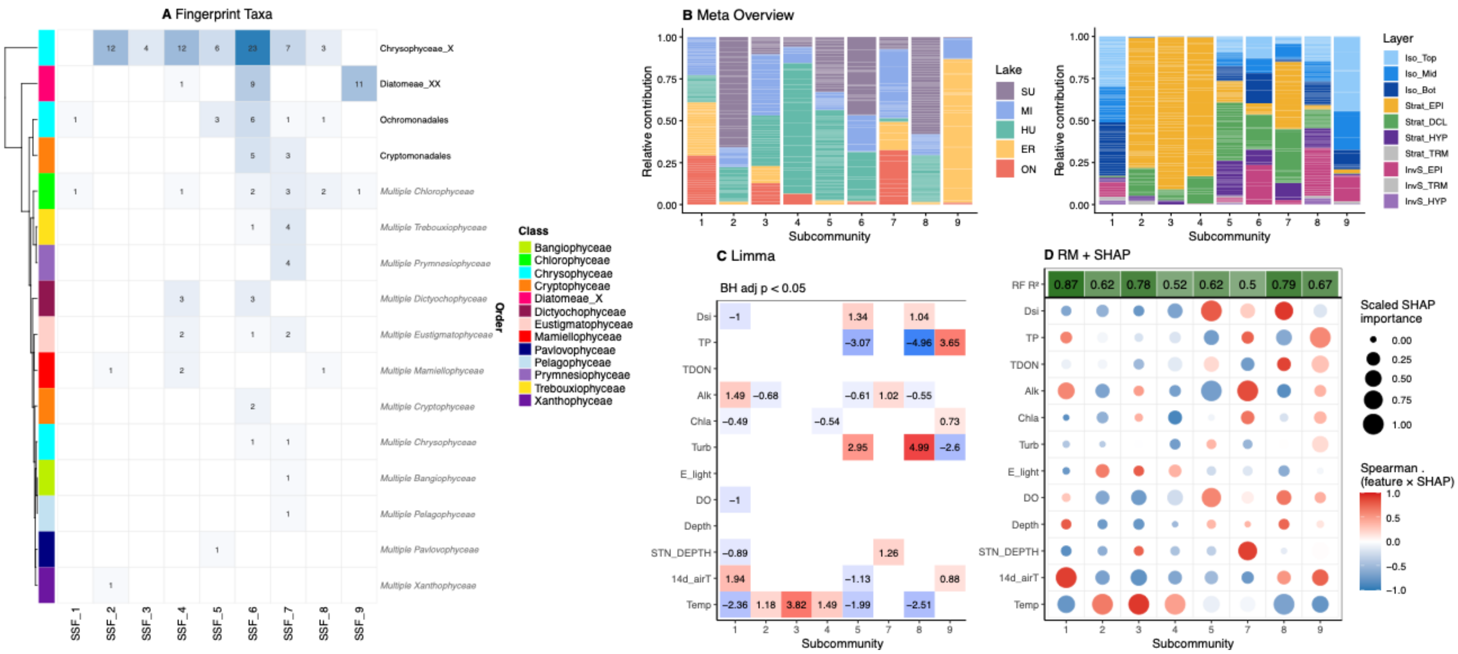
Supp. Fig 7. GoMDE distributions. (left) Distribution of all marker ASVs (n=2,044) by the number of SCs in which each ASV was significantly enriched. (right) log2 fold-change of the top 25 specialist ASVs per SC. Points are colored by phylum; pie charts above each SC show the phylum composition of all specialist ASVs for that topic.



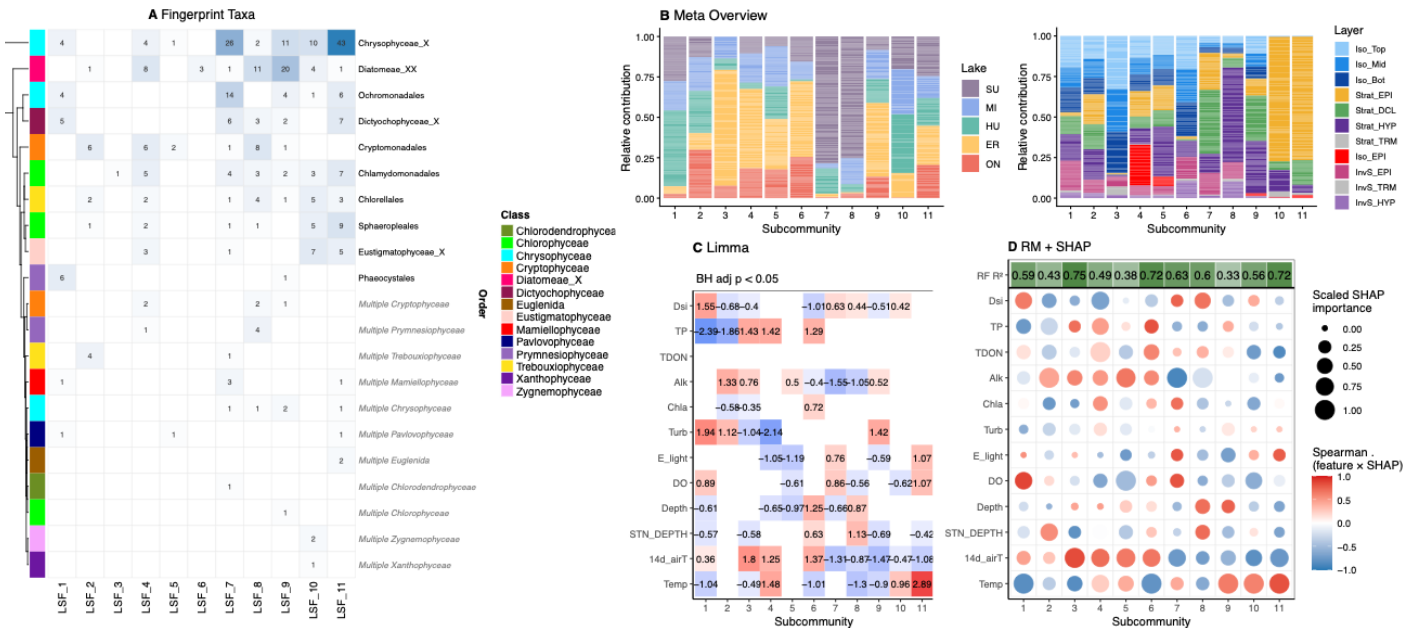
Supp. Fig. 8. Taxonomic identity and environmental associations of free-living prokaryote (FL) subcommunities. (A) Fingerprint taxa identified by GoM-DE discriminant analysis. Each column represents one subcommunity (SC) and each row represents a bacterial order. Numbers indicate the count of differentially enriched ASVs (BH-adjusted $p < 0.05$) from that order in that SC, colored by phylum. SC-specific ASVs (unique to one SC) are shown in bold; orders appearing in italic represent multi-phylum groupings. (B) Metadata overview showing the relative contribution of each SC across samples, summarized by lake identity (left) and water column layer (right). (C) Limma linear model results showing significant associations (BH-adjusted $p < 0.05$) between SC proportions and environmental variables. Numbers indicate standardized effect sizes; red indicates positive association and blue indicates negative association. (D) Random forest predictive importance (RF R^2 shown in header) and SHAP-based environmental driver analysis. Circle size reflects scaled SHAP importance and color reflects the Spearman correlation between feature value and SHAP value, indicating direction of effect.



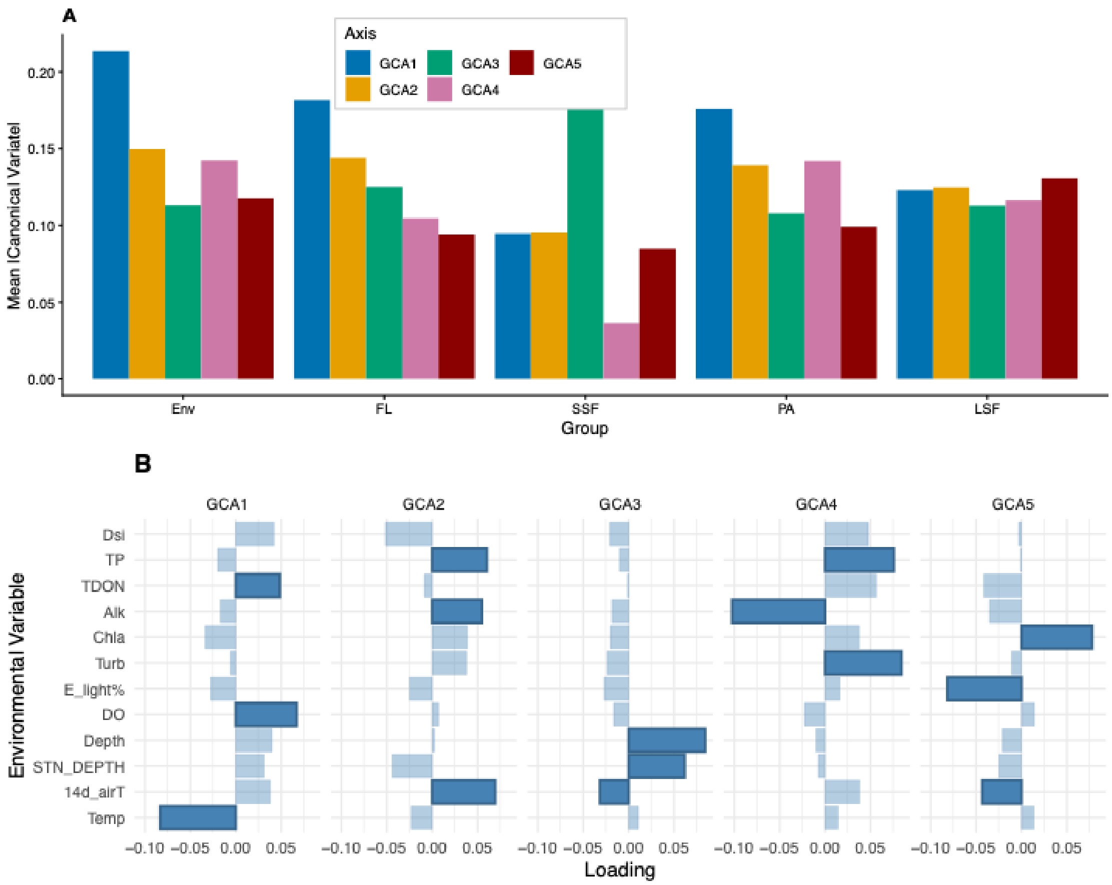
Supp. Fig. 9. Taxonomic identity and environmental associations of particle-associated prokaryote (PA) subcommunities. (A) Fingerprint taxa identified by GoM-DE discriminant analysis. Each column represents one subcommunity (SC) and each row represents a bacterial order. Numbers indicate the count of differentially enriched ASVs (BH-adjusted $p < 0.05$) from that order in that SC, colored by phylum. SC-specific ASVs (unique to one SC) are shown in bold; orders appearing in italic represent multi-phylum groupings. (B) Metadata overview showing the relative contribution of each SC across samples, summarized by lake identity (left) and water column layer (right). (C) Limma linear model results showing significant associations (BH-adjusted $p < 0.05$) between SC proportions and environmental variables. Numbers indicate standardized effect sizes; red indicates positive association and blue indicates negative association. (D) Random forest predictive importance (RF R^2 shown in header) and SHAP-based environmental driver analysis. Circle size reflects scaled SHAP importance and color reflects the Spearman correlation between feature value and SHAP value, indicating direction of effect.



Supp. Fig. 10. Taxonomic identity and environmental associations of small size fraction chloroplast (SSF) subcommunities. (A) Fingerprint taxa identified by GoM-DE discriminant analysis. Each column represents one subcommunity (SC) and each row represents a phytoplankton order. Numbers indicate the count of differentially enriched ASVs (BH-adjusted p < 0.05) from that class in that SC, colored by taxonomic class. SC-specific ASVs (unique to one SC) are shown in bold; classes appearing in italic represent multi-class groupings. (B) Metadata overview showing the relative contribution of each SC across samples, summarized by lake identity (left) and water column layer (right). (C) Limma linear model results showing significant associations (BH-adjusted p < 0.05) between SC proportions and environmental variables; SSF_6 is excluded as no associations passed the significance threshold. Numbers indicate standardized effect sizes; red indicates positive association and blue indicates negative association. (D) Random forest predictive importance (RF R² shown in header) and SHAP-based environmental driver analysis; SSF_6 is excluded (RF R² = 0.30, the lowest predictability observed across all SCs in the study), suggesting this subcommunity is not strongly structured by the measured environmental variables. Circle size reflects scaled SHAP importance and color reflects the Spearman correlation between feature value and SHAP value, indicating direction of effect.



Supp. Fig. 11. Taxonomic identity and environmental associations of large size fraction chloroplast (LSF) subcommunities. (A) Fingerprint taxa identified by GoM-DE discriminant analysis. Each column represents one subcommunity (SC) and each row represents a phytoplankton order. Numbers indicate the count of differentially enriched ASVs (BH-adjusted p < 0.05) from that order in that SC, colored by class. SC-specific ASVs (unique to one SC) are shown in bold; orders appearing in italic represent multi-class groupings. (B) Metadata overview showing the relative contribution of each SC across samples, summarized by lake identity (left) and water column layer (right). (C) Limma linear model results showing significant associations (BH-adjusted $p < 0.05$) between SC proportions and environmental variables. Numbers indicate standardized effect sizes; red indicates positive association and blue indicates negative association. (D) Random forest predictive importance (RF R^2 shown in header) and SHAP-based environmental driver analysis. Circle size reflects scaled SHAP importance and color reflects the Spearman correlation between feature value and SHAP value, indicating direction of effect.



Supp. Fig. 12. Group Compositional Analysis (GCA) of subcommunity distributions across all four biological blocks. (A) Mean absolute canonical variate scores for each GCA axis (GCA1–GCA5) across the five variable groups — environmental variables (Env), free-living prokaryotes (FL), small size fraction chloroplasts (SSF), particle-associated prokaryotes (PA), and large size fraction chloroplasts (LSF) — reflecting the relative contribution of each group to each axis. (B) Environmental variable loadings for each GCA axis. Bar length reflects the magnitude of each variable's loading and direction indicates positive or negative association.